

Pokhara University

Faculty of Sciences and Technology

Scholarship Entrance Examination – 2017

School of Engineering (SOE), Pokhara

Name of Student:

FM: 150

Exam Roll No:

Time: 3 hrs

Group 'A' Mathematics

- In any triangle $\frac{\sin(B-C)}{\sin(B+C)} =$
 - $\frac{b-c}{b+c}$
 - $\frac{b-c}{a}$
 - $\frac{\Delta}{bc}$
 - $\frac{b^2-c^2}{a^2}$
- The value of $\frac{r_1 r_2 r_3}{r}$
 - R^2
 - S^2
 - r^2
 - Δ^2
- The principal value of $\cos^{-1} \left\{ \cos \left(\frac{7\pi}{6} \right) \right\}$
 - $7\pi/6$
 - $5\pi/6$
 - $\pi/3$
 - none
- $\sum_{r=1}^{360} \sin x^0$
 - 1
 - 0
 - 1
 - none
- The value of $\sin^2 25 + \sin^2 65 =$
 - 1
 - 0
 - $\frac{1}{2}$
 - 1
- If $\sin\theta + \cos\theta = 1$ then the value of $\sin 2\theta =$
 - 0
 - 1
 - 1
 - $\frac{1}{2}$
- The period of $\sin 4x + \cos 4x$ is
 - π
 - $\frac{\pi}{2}$
 - $\frac{\pi}{3}$
 - $\frac{\pi}{6}$
- If $\tan^{-1} x, \tan^{-1} y, \tan^{-1} z$ are in A.P. and x, y, z are in A.P. then
 - $x + y + z = 0$
 - $x + y = z$
 - $x = y = z$
 - $x \neq y \neq z$

- Available on notes.arpanadhikari7.com

19. If $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ then $A^5 =$
- a) 5 A b) 10 A c) 16 A d) 32 A
20. The equation $|z + 1| = \sqrt{3} |z - 1|$ represents
- a) circle b) parabola c) hyperbola d) ellipse
21. The additive inverse of complex number $z = (a, b)$ is
- a) $(-a, -b)$ b) $(a, -b)$ c) (b, a) d) $(b, -a)$
22. The ratio of $\log_{10} x$ in x is
- a) $0.4343 : 1$ b) $0.301 : 1$ c) $2.30 : 1$ d) none
23. The coefficient of X^{20} in the expansion of $(1 + 3x + 3x^2 + x^3)^{15}$ is
- a) ${}^{45}C_{20}$ b) ${}^{45}C_{15}$ c) ${}^{20}C_{15}$ d) none
24. If A, G, H denote respectively then A.M., G.M. and H.M. between two unequal positive integers then
- a) $G^2 = G/A$ b) $H^2 = G/A$ c) $A^2 = G/H$ d) $A = G^2/H$
25. 5th term of the series 2, 4, 8, 16 is
- a) 16 b) 32 c) 64 d) 40
26. The system of equations $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$ and satisfies $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ then the number of solutions are
- a) unique b) infinite many c) two solutions d) no solutions
27. In ΔABC , $\operatorname{cosec} A =$
- a) $\frac{abc}{R}$ b) $\frac{bc}{\Delta}$ c) $\frac{bc}{2\Delta}$ d) $\frac{abc}{2s}$
28. The value of $\frac{r_1 r_2 r_3}{r} =$
- a) R^2 b) S^2 c) r^2 d) Δ^2
29. Derivative of an even function $f(x)$ is
- a) even function b) odd function
c) neither even nor odd d) none
30. The expression $\sin^2 \theta = \frac{x^2 + y^2}{2xy}$ is positive if
- a) $x = -y$ b) $x = y$ c) $x > y$ d) $x < y$

31. The function $f(x) = |x|$ is
 a) continuous and derivable at $x = 0$
 b) neither continuous nor derivable at $x = 0$
 c) continuous but not derivable at $x = 0$
 d) none
32. $f(x) = \frac{|x-a|}{x-a}$ when $x \neq a = 1$ when $x = a$, Then
 a) f is continuous every where
 b) f is continuous at $x = a$
 c) f has limit of at $x = a$
 d) limit of f does not exist at $x = a$
33. If the sum of the unit vectors is a unit vector then the magnitude of their difference is
 a) 1
 b) $\sqrt{3}$
 c) 3
 d) $2\sqrt{3}$
34. From 8 persons, in how many ways can a selection of 4 be made when two particular persons are always included?
 a) 8C_2
 b) 8C_4
 c) 6C_2
 d) 6C_4
35. Number of ways in which 4 boys and 3 girls can be arranged alternately in a row containing seven seats is
 a) $6!$
 b) $7!$
 c) $4!3!$
 d) $3!4!$
36. α and β are the roots of the equation $2x^2 - 15x + 9 = 0$ then the value of $\frac{1}{\alpha} + \frac{1}{\beta} =$
 a) $3/5$
 b) $2/9$
 c) $1/9$
 d) $/3$
37. If f is function from the set A to set B then
 a) A is called the domain of f
 b) B is called the domain of A
 c) A is called pre image of B
 d) B is called range of f
38. The value of $\log 6$ is equal to
 a) $\log 2 + \log 3$
 b) $\log 1 + \log 3 - \log 2$
 c) $\log 1 \cdot \log 2 \cdot \log 3$
 d) $\log(1+2) + \log 3$
39. If $y = (\sin x)^x$ then $\frac{dy}{dx} =$
 a) $y(x \cot x + \log x)$
 b) $y(x \cot x + \log \sin x)$
 c) $y(\log \sin x + \log x)$
 d) $y\left(\frac{\log \sin x}{x} - \cot x\right)$

40. If $\int f(x) dx = f(x)$ then

a) $f(x) = x$

b) $f(x) = \text{constant}$

c) $f(x) = ex$

d) $f(x) = 0$

Group 'B' Chemistry

41. The no. of electrons in $[_{19}\text{K}^{40}]^{-1}$ is

a) 18

b) 19

c) 20

d) 40

42. The standard entropies of $\text{H}_{2(g)}$, $\text{I}_{2(s)}$ and $\text{H}_{I(g)}$ are 130.6, 116.7 and 206.3 J-K-1-mol-1 respectively. The change in standard entropy in the reaction $\text{H}_{2(g)} + \text{I}_{2(s)} \rightarrow 2\text{HI}_{(g)}$ is

a) $165.9 \text{ J} - \text{K}^{-1} - \text{mol}^{-1}$

b) $170.5 \text{ J} - \text{K}^{-1} - \text{mol}^{-1}$

c) $175.6 \text{ J} - \text{K}^{-1} - \text{mol}^{-1}$

d) $180.7 \text{ J} - \text{K}^{-1} - \text{mol}^{-1}$

43. The half – life period depends upon

a) Rate constant

b) duration of the reaction

c) initial concentration of reactants

d) final concentration of products

44. The rate of reaction, which does not involve gases, does not depend upon

a) Catalyst

b) pressure

c) temperature

d) concentration

45. In Bohr's model, atomic radius of the first orbit is r . The radius of the third orbit is

a) $r/3$

b) r

c) $3r$

d) $9r$

46. If dilute HCl is added to the equilibrium reaction at constant temperature, then

a) Equilibrium constant will increase

b) concentration of CH_3COO^- will increase

c) equilibrium constant will decrease

d) concentration of CH_3COO^- will decrease

47. In acidic medium, the equivalent weight of KMnO_4 is equal to

a) Its molecular weight

b) half its molecular weight

c) one-sixth its molecular weight

d) one-fifth its molecular weight

48. Which of the following law directly explains the law of conservation of mass?
 a) Hund's rule
 b) Dalton's law
 c) Avogadro's law
 d) Berzelius hypothesis
49. The uncertainty in the momentum of a particle is $6 \times 10^{-2} \text{ kg-m-s}^{-1}$. The uncertainty in its position is
 a) $4.4 \times 10^{-14} \text{ m}$
 b) $6.8 \times 10^{-21} \text{ m}$
 c) $7.8 \times 10^{-30} \text{ m}$
 d) $8.8 \times 10^{-34} \text{ m}$
50. Oxygen readily dissolves in:
 a) H_2O
 b) Turpentine oil
 c) CS_2
 d) Ammonia
51. Ozone turns starch iodide paper
 a) Green
 b) Blue
 c) Red
 d) Black
52. A metal never found in free state is:
 a) Au
 b) Ag
 c) Cu
 d) Zn
53. Granite is a/an
 a) Alloy
 b) Amalgam
 c) Rock
 d) None
54. PCl_5 is kept in well stopped bottle because?
 a) It is volatile
 b) It reacts with oxygen
 c) It reacts with moisture
 d) It is explosive
55. In NO_3^- ion the number of lone pairs and bond pairs:
 a) 2, 2
 b) 3, 1
 c) 1, 3
 d) 4, 0
56. The absorption of hydrogen by palladium is called:
 a) Hydratium
 b) Reduction
 c) Occlusion
 d) Hydrogenation
57. H_2O is liquid while H_2S is gas because oxygen has
 a) high electro negativity
 b) low electro negativity
 c) high ionization energy
 d) low ionization energy
58. The arrangement of Na, Rb, K, Cs in the increasing order of atomic radius, is
 a) $\text{Na} < \text{K} < \text{Rb} < \text{Cs}$
 b) $\text{K} < \text{Na} < \text{Cs} < \text{Rb}$
 c) $\text{Na} < \text{Cs} < \text{K} < \text{Rb}$
 d) $\text{Rb} < \text{K} < \text{Cs} < \text{Na}$

59. Which of the following is not an amphoteric substance?
a) NH_3 b) H_2O c) HCO_3^- d) HNO_3
60. Maximum freezing point falls in
a) Camphor b) Naphthalene c) Benzene d) Water
61. If 480 ml of 1.5 M solution and 520 ml of 1.2 M solution is mixed, then molarity of the final solution is
a) 1.2 M b) 1.34 M c) 1.50 M d) 2.70 M
62. Which one of the following can be classified as a Bronsted base?
a) NO_3^- b) H_3O^+ c) NH_4^+ d) CH_3COOH
63. BF_3 is acid according to
a) Arrhenius b) Lewis
c) bronsted and lowery d) All
64. If solubility product of HgSO_4 is 6.4×10^{-5} , then its solubility is
a) 8×10^{-3} b) 8×10^{-6} c) 6.4×10^{-5} d) 6.4×10^{-3}
65. Which is not a colligative property?
a) Freezing point b) Lowering of vapour pressure
c) Depression of freezing point d) Elevation of boiling point
66. The triple point of water is
a) 172 K b) 273 K c) 298 K d) 373 K
67. The volume of carbon dioxide gas evolved at S.T.P. by heating 7.3 g of $\text{Mg}(\text{HCO}_3)_2$ will be
a) 1120 ml b) 1640 ml c) 2240 ml d) 3340 ml
68. The molecular weight of methane is 16. If the pressure of this gas changes from 1 atm to 2 atm, then volume occupied by 1.6g of this gas will be
a) 0.56 litre b) 1.12 litre c) 2.24 litres d) 4.48 litres
69. A layer of coke is spread over bauxite during extraction of aluminium. This acts as a/an
a) Flux
b) Slag to remove impurities
c) Reducing agent
d) Insulation and does not allow heat to escape

70. Borax is prepared by treating colemanite with
 a) NaNO_3 b) NaCl c) Co_3O_4 d) NaHCO_3
71. Strength of Halogen acids varies as:
 a) $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$ b) $\text{HF} > \text{HBr} > \text{HCl} > \text{HI}$
 c) $\text{HF} = \text{HCl} > \text{HBr} = \text{HI}$ d) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$
72. Iodine is placed in two liquids C_6H_6 and H_2O .
 a) It dissolves more in C_6H_6 b) It dissolves more in H_2O
 c) Dissolves equally in both d) Does not dissolve in both
73. Percentage of lead in pencil is
 a) 50% b) 20% c) 10% d) 0%
74. Prussian blue is:
 a) $\text{Fe}_2 [\text{Fe} (\text{CN})_6]$ b) $\text{Fe}_3 [\text{Fe} (\text{CN})_6]$
 c) $\text{Fe} [\text{Fe} (\text{CN})_6]_3$ d) $\text{Fe}_4 [\text{Fe} (\text{CN})_6]_3$
75. Carborundum is:
 a) Ca_2C b) CaC_2 c) SiC d) none
76. Equilibrium constant for the reaction $2\text{A}_{(\text{g})} \rightleftharpoons \text{B}_{(\text{g})} + \text{C}_{(\text{g})}$ at 780 K and 10 atm pressure is 3.52. The equilibrium constant of this reaction at 780K and 20 atm pressure is
 a) 3.52 b) 5.72 c) 11.44 d) 28.6
77. The aq. solution of sodium cyanide is
 a) Acidic b) Amphoteric c) Basic d) Neutral
78. The oxidation number of carbon in CH_2O is
 a) -2 b) +2 c) 0 d) +4
79. Which of the following is the strongest oxidizing species?
 a) Na^+ b) Li^+ c) Ba^{2+} d) Mg^{2+}
80. Intramolecular H-bonding is present in
 a) Water b) ammonia
 c) o-nitrophenol d) hydrogen chloride

Group 'C' Physics

81. The resultant of two vectors of equal magnitudes is equal to the magnitude of either. The angle between the vector
 a) 60° b) 90° c) 120°

82. The ratio of angular speed of minute-hand and hour-hand of a watch is
a) 1:6 b) 6:1 c) 1:12 d) 12:1
83. If the radius of earth contracts to half of its present value, the length of the day will be
a) 6 h b) 12 h c) 24 h d) 48 h
84. On moving from the equator to the poles value of g
a) decreases b) increases
c) decreases up to latitude 45° d) same
85. With rise in temperature the Young's Modulus
a) increases b) decreases
c) remains unchanged d) changes erratically
86. A sphere rolls down an inclined plane without sliding. Then frictional force on the sphere act
a) Up the inclined plane parallel to the plane
b) Down the inclined plane parallel to the plane
c) Vertically upward
d) Vertically downward
87. When two tubes of different diameters are dipped vertically, the rise of liquid is
a) same in both tubes
b) more in tube of larger diameter
c) more on tube with smaller diameter
d) none of the above
88. In Milikan's drop experiment a small spherical oil drop of radius r is making in medium of density ρ with an instantaneous speed v . The viscous force acting on the drop is
a) $6\pi\eta rv$ b) ηrv c) $6\eta/rv$ d) $6\eta rv/\pi$
89. A piece of ice is floating in jar containing water when the ice melts, then level of water
a) rises b) falls
c) remains constant d) may rise or fall

90. The gas thermometer are more sensitive than liquid thermometer because
- gas expands more than liquids
 - gas are easily obtained
 - gases are much lighter
 - gases don't easily change their state
91. A copper disc has a hole. If the disc is heated the size of hole
- increases
 - decreases
 - remains unchanged
 - first decreased and then increases
92. The temperature of gas is measure of
- the average K.E. of the gas molecules
 - the average P.E. of the gas molecules
 - the average distance between the molecules of gas
 - the size of the molecules of the gas
93. The average energy of the molecules of mono atomic gas at temperature T is
- $\frac{1}{2}KT$
 - KT
 - $\frac{3}{2}KT$
 - $\frac{5}{2}KT$
94. An iron block cools from 100°C to 90°C in time t_1 ; 90°C to 80°C in time t_2 and 80°C to 70°C in time t_3 . Which of the following relations is correct?
- $t_1 > t_2 > t_3$
 - $t_1 < t_2 < t_3$
 - $t_1 = t_2 = t_3$
 - $t_3 = \frac{t_1 + t_2}{2}$
95. The area under the indicator diagram gives
- heat gained or lost by the system
 - work done on the system or by the system
 - average kinetic energy of the particles of the system
 - none of these
96. Two bodies are at temperatures 27°C and 327°C . The heat energy radiated by them will be in the ratio
- $\frac{1}{2}$
 - $\frac{1}{8}$
 - $\frac{1}{16}$
 - $(27/327)^{\circ}$
97. The relation between phase difference and path difference is
- $\Delta\phi = \frac{2\pi}{\lambda} \Delta x$
 - $\Delta\phi = 2\pi\lambda \Delta x$
 - $\Delta\phi = \frac{2\pi\lambda}{\Delta x}$
 - $\Delta\phi = \frac{\pi\Delta x}{\lambda}$

98. Both length and diameter of the stretched string are doubled. If earlier the frequency of the fundamental note was n , now the frequency will be equal to
 a) $4n$ b) n c) $n/4$ d) none
99. Combination of two tones that gives pleasing effect is termed as
 a) beat b) concord c) chord d) harmony
100. Which of the following is not the case with the images formed by a plane mirror?
 a) it is erect
 b) it is virtual
 c) it is diminished
 d) it is at the same distance as the object
101. The focal length of a glass lens is
 a) less for red and more for violet rays
 b) less for violet and more for red rays
 c) same for all colours
 d) a and b may be true depending on nature of glass
202. The angle of prism is 8° , it gives a deviation of 4° , the refractive index of material is
 a) 1.2 b) 1.33 c) 1.5 d) 1.45
103. The amount of light reaching the retina of the eye is controlled by
 a) cornea b) pupil
 c) iris d) ciliary's muscle
104. For a long cylindrical source of light, the illuminance E on a screen varies with distance r as
 a) $E \propto r^2$ b) $E \propto r^{-1}$ c) $E \propto r^0$ d) $E \propto r^{-2}$
105. For achromatic combination of lenses if we use two lenses of focal lengths f and f' , dispersive powers w and w' respectively then
 a) $w = w_0$, $w' = 2w_0$, $f' = 2f$ b) $2 = w_0$, $w' = 2w_0$, $f = -2f$
 c) $w = w_0$, $w' = 2w_0$, $f' = 2f$ d) $w = w'$, $w' = 2w'$, $f = -f/2$
106. A ray of light is incident on a glass plate at 60° . The reflected and refracted rays are found to be mutually perpendicular. The refractive index of glass is
 a) 2 b) 1.73 c) 1.50 d) 1.15

107. The magnitude of the two charges is doubled and the distance of their separation is also doubled. The electrostatic force between them will
- be halved
 - be doubled
 - become four times
 - remain unchanged
108. A capacitor works in
- AC circuits
 - DC circuits
 - both AC and DC circuits
 - neither AC nor DC circuits
109. If P, Q, R and S are resistance of the four arms of the Wheatstone bridge, then the bridge is most sensitive when
- $P = Q = R = S$
 - $P = Q$ and $R = S$
 - $P + Q = R + S$
 - $P - Q = R - S$
110. You are given three bulbs of 25, 40 and 60 watts, which of them has lowest resistance
- 25w bulb
 - 40w bulb
 - 60w bulb
 - none
111. A solenoid is 1m long and it has 4250 turns. If a current of 5A is flowing through it, what is the magnetic field at the centre?
- $5.4 \times 10^{-2} \text{ wb/m}^2$
 - $2.7 \times 10^{-2} \text{ wb/m}^2$
 - $1.25 \times 10^{-2} \text{ wb/m}^2$
 - $0.675 \times 10^{-2} \text{ wb/m}^2$
112. Scalar potential on the equatorial line of a bar magnet is
- zero
 - less than zero
 - more than zero and finite
 - infinite
113. The current and voltage through inductance differ in phase by
- $\pi/4$
 - $\pi/2$
 - $3\pi/4$
 - π
114. When an electron moves in a transverse magnetic field, the path becomes
- straight line
 - circular
 - parabolic
 - elliptical
115. Which of the following phenomenon support the quantum nature of light?
- interference
 - diffraction
 - polarization
 - Compton effect

116. Rutherford's α -particle scattering experiment gave experimental information about
- charge of α -particle
 - size of atom
 - size of nucleus
 - the force between α -particle & proton in the nucleus
117. The dimension of Rydberg constant are
- $[M^0L^{-1}T^0]$
 - $[M^0L^{-1}T]$
 - $[M^0L^0T^{-2}]$
 - $[MI^{-7}T]$
118. The mass defect for the nucleus of helium is 0.0303 amu. What is the binding energy per nucleon for helium in MeV?
- 1
 - 4
 - 7
 - 27
119. What is the number of junction in a semiconductor diode?
- one
 - two
 - three
 - no
120. A particle is in motion for its motion in certain time displacement covered is
- Always equal to distance covered
 - Always less than distance covered
 - Greater than distance covered
 - Less than or equal to distance covered

Group 'D' English

121. The students hope the exam.
- to pass
 - pass
 - to passing
 - passing
122. He looks as if he sick.
- had been
 - N
 - was
 - were
123. He postponed to Lumbini
- going
 - to going
 - gone
 - to go
124. She said, "Man is mortal." Its direct speech is
- She said that man was mortal
 - She told man is mortal
 - She said man is mortal
 - She said that man is mortal

125. The passive voice of 'let me do it' is
 a) It should be done by me b) Let it be done by me
 c) Let I should do it. d) Let it be done by
126. Which of the following is compound sentence?
 a) The cat purred
 b) Tee can be rude at times but she is nice girl
 c) The big dog barked whenever I knocked on the door
 d) Became life is complex, we cannot remain at rest
127. 'Stop it.' It is.
 a) Imperative sentence b) Interrogative sentence
 c) Assertive sentence d) Exclamatory sentence
128. Either you or your brother responsible
 a) is b) are c) were d) have
129. The antonym of words 'stringent' is
 a) strict b) puritanical c) flexible d) vital
130. The synonyms of 'carnal' is
 a) sexual b) stormy c) hasty d) spiritual
131. One who beives in god is
 a) protocol b) atheist c) polyglot d) theist
132. Study of man kind
 a) calligraphy b) entomology
 c) anthropology d) philanthropy
133. The love of beauty
 a) Aesthetic b) Aristocracy c) Acumen d) Anthropolog
134. The family is sitting the dining table.
 a) in b) on c) at d) by
135. Nepal had victory its rival team
 a) by b) over c) of d) with
136. He is suffering typhoid
 a) of b) from c) off d) with
137. 'To break the ice' refers to
 a) to end the hostility b) to end up partnership
 c) to start quarrelling d) to start a conversation

138. Which of the following is correct?

- a) I want to buy the new jacket but it is too expensive
- b) I want to by the new jacket but it is too expensive
- c) I want to buy the new jacket, but it is too expensive
- d) I want to buy the new jacket, but it is too expensive

139. Which of the following sentences is correctly punctuated?

- a) That's unbelievable!
- b) Charles book is very good:
- c) He had just one fault: an enormous ego
- d) All these

140. The word 'economy' has its primary stress on its syllable.

- a) 1st
- b) 2nd
- c) 3rd
- d) 4th

141. The basic sentence pattern for "Students don't vote." is.

- a) S + V
- b) S + V + O
- c) S + V + Sub complement
- d) S + V + obj commitment

142. He since last Sunday

- a) has worked
- b) have worked
- c) hasn't worked
- d) haven't worked

143. She the work by tomorrow

- a) will finish
- b) will have finished
- c) finished
- d) is finishing

144. He worked hard so that he good

- a) will be earning
- b) may earn
- c) earned
- d) would earn

145. 'Solar' is for 'Sun.' is for 'Galaxy.'

- a) Galactar
- b) Galactic
- c) Galaxian
- d) All of above

146. It is more expensive to travel on Friday, I will leave on Sunday Moring. The underlined word is

- a) adverb
- b) adjective
- c) verb
- d) noun

147. Which of the following contains [f] sound?

- a) voice
- b) compressive
- c) proof
- d) vex

148. Which of the following is true?

- a) I can help you: I speak a little French
- b) I can help you: I speak little French
- c) Both
- d) None

149. Please excuse being late.

- (a) me
- (b) my
- (c) mine
- (d) myself

150. Unable to pay one's debts

- (a) indigent
- (b) beggar
- (c) debtor
- (d) insolvent

Answer Key

1. (d)	2. (b)	3. (b)	4. (b)	5. (d)	6. (a)	7. (b)
8. (c)	9. (a)	10. (d)	11. (d)	12. (b)	13. (d)	14. (c)
15. (a)	16. (d)	17. (d)	18. (b)	19. (c)	20. (b)	21. (a)
22. (a)	23. (a)	24. (d)	25. (b)	26. (b)	27. (d)	28. (b)
29. (b)	30. (b)	31. (c)	32. (d)	33. (b)	34. (c)	35. (c)
36. (d)	37. (a)	38. (a)	39. (b)	40. (c)	41. (c)	42. (a)
43. (a)	44. (b)	45. (d)	46. (d)	47. (d)	48. (b)	49. (d)
50. (b)	51. (b)	52. (d)	53. (c)	54. (c)	55. (d)	56. (c)
57. (a)	58. (a)	59. (d)	60. (a)	61. (b)	62. (a)	63. (b)
64. (a)	65. (a)	66. (b)	67. (a)	68. (b)	69. (d)	70. (c)
71. (d)	72. (a)	73. (d)	74. (d)	75. (c)	76. (a)	77. (c)
78. (c)	79. (d)	80. (c)	81. (c)	82. (d)	83. (a)	84. (b)
85. (b)	86. (a)	87. (c)	88. (a)	89. (c)	90. (a)	91. (a)
92. (a)	93. (c)	94. (a)	95. (b)	96. (b)	97. (a)	98. (c)
99. (d)	100. (c)	101. (b)	102. (c)	103. (c)	104. (b)	105. (b)
106. (b)	107. (d)	108. (a)	109. (a)	110. (c)	111. (b)	112. (a)
113. (b)	114. (b)	115. (d)	116. (c)	117. (a)	118. (c)	119. (a)
120. (d)	121. (a)	122. (c)	123. (a)	124. (d)	125. (b)	126. (b)
127. (a)	128. (a)	129. (c)	130. (a)	131. (d)	132. (c)	133. (a)
134. (c)	135. (b)	136. (b)	137. (d)	138. (d)	139. (d)	140. (b)
141. (a)	142. (c)	143. (b)	144. (b)	145. (b)	146. (a)	147. (c)
148. (a)	149. (b)	150. (a)				
